System Validation Report

Railway Reservation System - Team 15

1. Purpose and scope

This report records the manual system and acceptance validation of the Railway Reservation System through the command-line interface (GitHub Issue #23). It is based on the full execution of 31 Oct 2026 documented in docs/SYSTEM_ACCEPTANCE_TEST_REPORT.md, with screenshot evidence in docs/evidence/issue-23/.

Field	Value
Project	Railway Reservation System (RRS)
Team	Team 15 (TEAM15-RRS)
Repository	TEAM15-RRS/se-miniproject-railway
Code version tested	Branch main, commit 7ff1f9c (3 Nov 2026)
Report date	11 November 2026
Test plan	Software Test Plan v1.0 (16 Sep 2026)
Requirements	SRS v1.0 (13 Sep 2026)
Execution date	31 October 2026
Environment	Google Colab, Linux, C++17 build (build/rrs)

2. Summary of results

Result	Count	Share
PASS	17	63%
PARTIAL (evidence does not show the full acceptance condition)	3	11%
FAIL	3	11%
NOT EXECUTABLE through the CLI at the time	4	15%
Total RTM test cases	27	100%

The end-to-end workflow E2E-CLI-01 (register, login, search, book, PNR lookup, cancel, PNR lookup again) passed.

3. Test results by area

Test case	Requirement	Status	Note
TC-Reg-01	RRS-F-001 Registration	FAIL	Only username and password collected (DEF-02)
TC-Auth-01	RRS-F-002 Passenger login	PASS	
TC-Auth-02	RRS-F-003 Admin login	PASS	
TC-Search-01	RRS-F-010 Search by route	PASS	
TC-Search-02	RRS-F-011 Class availability	PASS	
TC-Search-03	RRS-F-012 RAC/WL indication	PASS	

Test case	Requirement	Status	Note
TC-Book-01	RRS-F-020 Passenger details	FAIL	No berth preference prompt (DEF-03)
TC-Book-02	RRS-F-021 Unique 10-digit PNR	PARTIAL	Uniqueness after reload not shown (covered by IT-05)
TC-Book-03	RRS-F-022 Seat decrement	PASS	72 to 71 seats
TC-Book-04	RRS-F-023 RAC/WL status	PASS	
TC-Book-05	RRS-F-024 Payment success/failure	PASS	
TC-Book-06	RRS-F-025 Receipt	PASS	
TC-Cancel-01	RRS-F-030 Cancel by PNR	PARTIAL	Other-user rejection not shown (covered by IT-11)
TC-Cancel-03	RRS-F-032 Promotion	PASS	
TC-Cancel-04	RRS-F-033 Cancelled record kept	PASS	
TC-PNR-01	RRS-F-034 PNR lookup	PASS	
TC-Admin-01	RRS-F-040 Add train	NOT EXEC.	No CLI menu option at the time
TC-Admin-02	RRS-F-041 Fare/quota update	PASS	Fare 745 to 800 reflected in booking
TC-Admin-03	RRS-F-042 Deactivate train	NOT EXEC.	No CLI menu option at the time
TC-Admin-04	RRS-F-043 Reports	NOT EXEC.	No CLI menu option at the time
TC-Data-01	RRS-F-050 Persistence	PASS	PNR 1000000003 after restart
TC-Data-02	RRS-F-051 Backup/restore	NOT EXEC.	No CLI menu option at the time
TC-Sec-01	RRS-SR-001 Hashed passwords	PASS	
TC-Sec-02	RRS-SR-002 Input masking	PASS	
TC-Sec-03	RRS-SR-003 Admin access control	PARTIAL	Direct invocation not shown (covered by unit test)
TC-Sec-04	RRS-SR-004 Admin log timestamp	FAIL	Timestamp 1970-12-12 (DEF-01)
TC-Sec-05	RRS-SR-005 Lockout	PASS	

4. Defect register

ID	Test	Defect	Requirement
DEF-01	TC-Sec-04	Admin action log records incorrect event timestamp (1970-12-12 22:54:40 instead of 31 Oct 2026)	RRS-SR-004
DEF-02	TC-Reg-01	Registration omits passenger name, age, gender and contact number	RRS-F-001
DEF-03	TC-Book-01	Booking does not collect berth preference	RRS-F-020

Insert the GitHub Issue number for each defect once the issue is created (label: bug, assigned to the module owner).

5. Retest status

Defect	Fix status	Retest
DEF-01 Admin log timestamp	[Update]	Repeat TC-Sec-04
DEF-02 Registration fields	[Update]	Repeat TC-Reg-01
DEF-03 Berth preference	[Update]	Repeat TC-Book-01
Four admin/data options (add train, deactivate, reports, backup)	[Update]	Repeat TC-Admin-01, 03, 04, TC-Data-02 if now in CLI menu

The code has changed since 31 Oct 2026: unit tests now cover add train, deactivate, admin reports and backup/restore. The four tests that were not executable through the CLI must be re-run if the menu now exposes those functions.

6. Supporting automated evidence

The latest run of the automated suite (141 unit, 12 integration, 1 performance test) passes in full, and covers the partial cases above (IT-05 PNR uniqueness after reload, IT-11 cancel by another user, TC_AdminAccess_RejectsPassengerActions). See the Unit Test Report.

7. Conclusion

17 of 27 acceptance cases passed, and the full passenger workflow from registration to cancellation works. Three requirement defects (DEF-01 to DEF-03) were found and need fixes and retests. The system is acceptable for the demonstrated core flows, but it is not fully ready until the defects are closed and the four pending admin/data tests are executed.